

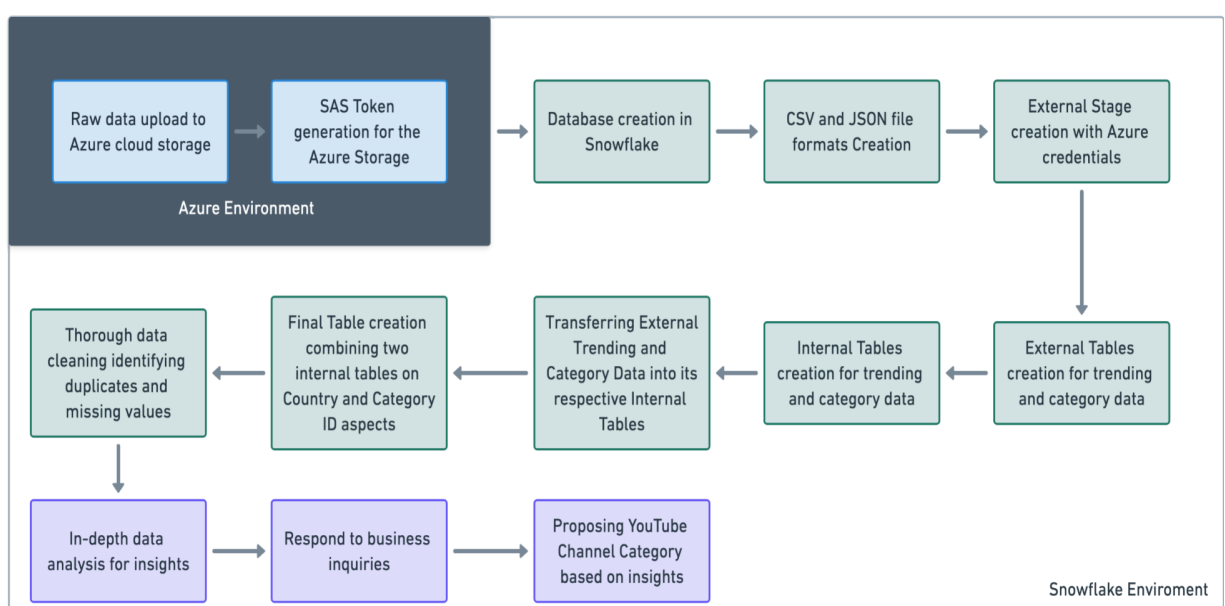
1. Introduction

The assignment focuses on analyzing a large dataset of trending YouTube videos from different regions over a long time period – August 2020 to April 2024. Implementing Microsoft Azure and Snowflake, the goal of the assignment is to demonstrate how to manage and analyze a large, raw, and semi-structured dataset in a Data Lakehouse architecture, including data ingestion and transformation, followed by data cleaning to ensure the dataset is consistent by identifying and resolving issues such as duplicates, missing values, and erroneous data entries. Subsequently, the aim is to derive meaningful insights and address certain queries concerning YouTube trends across different regions. Based on the analysis, identifying the optimal content strategy for launching a new YouTube channel is the ultimate objective.

1.2 Overview of Key Stages

- Uploading the folder of raw data for trending YouTube videos and metadata of the categories, onto the Azure cloud storage.
- Data ingestion and converting it to Snowflake tables.
- Data cleaning to ensure accuracy and consistency.
- Data analysis to derive significant insights and respond to particular business inquiries concerning regional trends in YouTube videos.

Figure 1: Flowchart of YouTube Trending Analysis in Data Lakehouse Architecture



2. Data Source Exploration

The top trending YouTube videos for ten regions – India (IN), USA (US), Great Britain (GB), Germany (DE), Canada (CA), France (FR), Brazil (BR), Mexico (MX), South Korea (KR), Japan (JP) – are recorded in this dataset. The YouTube API was used to gather the dataset, which includes daily data collection over a certain period, – from 2020-08-12 to 2024-04-15.

2. 1. Data Structure

2.1.1 Trending Data: Ten CSV files, one for each of the ten regions, make up the dataset in a folder for trending videos. Up to 200 videos are listed each day in each file, which provides information on the most popular videos in that area. The folder for trending data contained 10 CSV files, each containing 12 columns: VIDEO_ID, TITLE, PUBLISHEDAT, CHANNELID, CHANNELTITLE, CATEGORYID, TRENDING_DATE, VIEW_COUNT, LIKES, DISLIKES, COMMENT_COUNT, COUNTRY.

2.1.2 Category Data: Ten JSON files with metadata about the different video categories for every region are also included in a folder for category data, which allows for decoding the category information in the trending data.

3. Explanation of the Different Steps

3.1 Cloud Storage Set up and Data Upload

The raw datasets containing the CSV and JSON files have been uploaded to Azure cloud storage. To provide delegated access to the resources to be able to use in Snowflake, a SAS token has been generated in the Azure environment.

3.2 Database Creation and Data Ingestion

Once the data was uploaded to the Azure storage container, a Snowflake database named “assignment_1” was created, and an external stage was set up, pointing to the Azure storage location. The data was ingested into Snowflake as external tables, using appropriate file formats for CSV and JSON data.

Figure 2: Snippet of External Table for Trending Data

VALUE	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	TRENDING_DATE	VIEW_COUNT	LIKES	DISLIKES	COMMENT_COUNT
{ "c1": "uq5LQION3e", "c10": "105756", "c11": "139946", "c2": "안녕하세요 보겠습니다", "c3": "2020-08-09T09:32:48Z", "c4": "UCu9BCiGIE73LXZsKmoUjKw", "c5": "보컬 BK", "c6": "24", "c7": "2020-08-12T00:00:00Z", "c8": "5947503", "c9": "53326" }	uq5LQION3e	안녕하세요 보겠습니다	2020-08-09 09:32:48.000 Z	UCu9BCiGIE73LXZsKmoUjKw	보컬 BK	24	2020-08-12 00:00:00.000 Z	5947503	53326	105756	139946
{ "c1": "1-ZbZCHsHD0", "c10": "484", "c11": "3339", "c2": "부락토스의 계획 [총명명 프리뷰]", "c3": "2020-08-12T09:00:08Z", "c4": "UCRU8xVv4qTK5KCh90ntAgA", "c5": "총명명", "c6": "1", "c7": "2020-08-12T00:00:00Z", "c8": "963384", "c9": "28244" }	1-ZbZCHsHD0	부락토스의 계획 [총명명 프리뷰]	2020-08-12 09:00:08.000 Z	UCRU8xVv4qTK5KCh90ntAgA	총명명	1	2020-08-12 00:00:00.000 Z	963384	28244	494	3339

Figure 3: External Table for Category Data

VALUE	[] VALUE
{ "etag": "kBCr3I9kLHHU79W4Ip5196LDptI", "items": [{ "etag": "IfWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp28mDpdntcQ", "title": "Film & Animation" } }] }	{ "etag": "kBCr3I9kLHHU79W4Ip5196LDptI" , "items": [{ "etag": "IfWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp28mDpdntcQ", "title": "Film & Animation" } }] }

3.3 Data Transformation

The data needed to be transformed into a format for it to be analyzed after it was ingested. Internal tables have been created in Snowflake to store the trending video data and category metadata independently. Among the steps in the transformation process are:

- Transforming the data fields into the relevant forms.
- Obtaining Country codes from the filenames.
- Bringing the category metadata from several regions and normalizing it.

In the final transformation phase, the trending video data and the category information have been joined into a single table called “table_youtube_final.” To enable additional analysis, a unique identifier was appended to every record through the utilization of the ‘UUID_STRING()’ function.

Figure 4: Data in “table_youtube_final” limit to 10

ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	CATEGORY_TITLE	TRENDING_DATE	VIEW_COUNT	LIKES	DISLIKES	COMMENT_COUNT	COUNTRY
1ec77930-43e6-431e-ab53-40b46e7d6d3	KJ2ggFF-RE	Bonnet MC - HOLLYWOOD (Snippet)	2020-08-11	UC8thmH8d9tq2mX0t9tg	ChymeTV	10	Music	2020-08-12	573902	69319	970	3311	DE
0ba4f834-92f6-4b6d-8a6d-94f8a13865a	KQvYhOn7wZ	Nik hat heftige Probleme in Köln! 🤯👊 #1925 Köln 50667	2020-08-11	UC1nUg5AMJWPDSrv_vo7TAqA	Köln 50667	24	Entertainment	2020-08-12	381375	13837	435	886	DE
0a9e531b-6c0d-4b69-88f9-2785a3f16e8a	2bzn8b79LRc	Camper Tour 2020 - ROADTRIP durch Österreich Episode #2 AnaJohnson	2020-08-11	UCB8RY61tvanhKzceZdN1Cw	AnaJohnson	24	Entertainment	2020-08-12	142296	9480	144	364	DE
207ba020-fa8f-4d77-ba63-3de0713407d	Zv-3qheAMaM	Ich TESTE SHEN BIKINIS (by on haul) - UNMÖGLICH "net" (in shock) BYE	2020-08-12	UCcc0h46QpRcjMgS98nA	Erlach Marcel	24	Entertainment	2020-08-12	55640	3420	124	229	DE
25cb1cde-4c18-4b52-b909-2288630a310	7c9jQLneouJ	STATEMENT zu meinem BAN 🙏 Bald vielleicht PERMABAN...? Pappalatte Highlights	2020-08-11	UC8E8eD7mCocMaz7f4uK6FQ	Domo	24	Entertainment	2020-08-12	233899	25251	375	1051	DE
95de9286-22ed-4cbe-bd56-24a913b7603a	q7vrdQg7Kic	Bayern unterliegt abgezockten Mailändern: Inter Mailand - Leverkusen 2:1 UEFA Europa League DAZN	2020-08-10	UCN9qgOKJONcC3QUWHLuKw	DAZN UEFA Europa League	17	Sports	2020-08-12	623938	12770	357	1514	DE
bcd54962-2991-49e0-ab13-12bc7298791	85gth8tSE	GERBRAUCHTER Schrott - Die neue Masche?	2020-08-11	UC2H549Y2hdqkhtvz7dHmQ	AlexBeal	24	Entertainment	2020-08-12	249531	17199	304	812	DE
035eca3-4532-448a-be80-6943b8a92e1d	YECPrRks4	Erkenntest DU den Song? (mit unsympathischTV) - DAS FINALE!	2020-08-11	UCL5-Phm1_sIES7ZcYJRp5A	World Wide Wohnzimmer	24	Entertainment	2020-08-12	470201	43045	369	1719	DE
661d5646-5137-4bdc-821b-56c29479868d	q3VMMFHyg	Giannis Gets Ejected After Headbutting Mo Wagner	2020-08-12	UC9-QpMMvNPN5a10_Jyq7Ndw	Bleacher Report	17	Sports	2020-08-12	756814	8278	331	2441	DE
12587e40-5447-4e9e-8d37-049435a4305	q3QPHBdSrMl	Giannis' Ekinis aus dem Internet testen... "send help" Sonny Loops	2020-08-11	UCPUPECd8H8M7t9nBQ	Sonny Loops	26	Howto & Style	2020-08-12	154766	11261	673	733	DE

3.4 Data Cleaning

- The category titled 'Comedy' has duplicates when not considering the variable 'CATEGORYID'.
- Only one region appears to have the category "Nonprofits & Activism".
- The 'CATEGORYID' 29 corresponds to the records where 'CATEGORY_TITLE' is missing in 'table_youtube_final'.
- Using the 'CATEGORYID' from 'table_youtube_category', the correct 'CATEGORY_TITLE' has been substituted for the NULL values in 'CATEGORY_TITLE'.
- The video titled "Kala Official Teaser | Tovino Thomas | Rohith V S | Juvis Productions | Adventure Company" does not have a 'CHANNEL_TITLE' in the dataset.
- All records with the video ID "#NAME?", which are the invalid video records, have been deleted from 'table_youtube_final' resulting in the removal of 32,081 rows.
- Using the 'ROW_NUMBER()' function, duplicate records were found by comparing their 'VIDEO_ID', 'TRENDING_DATE', and 'COUNTRY'; the record with the highest 'view_count' was retained in the original table. The discovered duplicates were placed in a new table called 'table_youtube_duplicates'. 37,466 duplicate rows were subsequently removed from 'table_youtube_final'.
- Following the data cleaning procedure, 'table_youtube_final' has 2,597,494 rows in total.

3.5 Data Analysis

To provide targeted answers on engagement metrics, category trends, and video popularity, a number of queries have been conducted.

- **Top 3 most viewed videos for each country in the Gaming category for the trending_date = "2024-04-01":**

The most viewed top video in several countries (Brazil, Canada, Germany, France, Great Britain, and Mexico) is "DAGGER DUCHESS - New Tower Troop!" by Clash Royale, from Brazil, Canada, Germany, France, Great Britain with 4.9 million views. View counts for the top videos vary from roughly 1.2 million to 7.2 million, with Mexico having the highest view count at 7.2 million for the video "HAPPY WHEELS, PERO EN 2024 !! - Episodio 28" and Japan having the lowest at 1.2 million for the video titled "働いたことない男のスーパーマーケット経営"

Supermarket Simulator 』”. With 4.3 million views, Techno Gamerz’s “I BUILD MY NEW HOUSE | PALWORLD GAMPLAY #8” is the most popular video in India. Korea’s most viewed video is “‘HOT DEBUT’ 아일릿(ILLIT) - Magnetic #엠카운트다운 EP.835 | Mnet 240328 방송” with a view of 1.7 million. In the USA, “Confrontation - The Skibidi Saga 05” by Maxedy is the most-viewed video with 3.2 million views.

Figure 5: Table of Top 3 most viewed gaming videos by country for 2024-04-01

COUNTRY	TITLE	CHANNEL/TITLE	VIEW COUNT	RK
BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	IShowSpeed	2971782	2
BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
CA	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
CA	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3
DE	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
DE	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
DE	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: Warzone	Call of Duty	2311131	3
FR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
FR	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: Warzone	Call of Duty	2311131	2
FR	Clove Official Gameplay Reveal // VALORANT	VALORANT	2043592	3
GB	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
GB	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
GB	IShowSpeed - Monkey (Official Music Video)	IShowSpeed	2655688	3
IN	I BUILD MY NEW HOUSE PALWORLD GAMPLAY #8	Techno Gamerz	4298290	1
IN	I BECAME A TAXI DRIVER	Techno Gamerz	4064687	2
IN	Ye Baby Nahi Shaitaan Hai - The Baby in Yellow (Part 1)	Live Insaan	3967222	3
JP	働いたことない男のスーパーマーケット経営『Supermarket Simulator』	キヨ。	3153412	1
JP	働いたことない男がバイトを雇うスーパーマーケット経営『Supermarket Simulator』	キヨ。	2116115	2
JP	【崩壊：スターレイル】黄泉 キャラクターPV 「君の色」	崩壊：スターレイル	1206271	3
KR	‘HOT DEBUT’ 아일릿(ILLIT) - Magnetic #엠카운트다운 EP.835 Mnet 240328 방송	Mnet K-POP	1678685	1
KR	(여자)아이들 - 나는 아픈 건 뭐 짝해이니까 #엠카운트다운 EP.834 Mnet 240321 방송	Mnet K-POP	1337298	2
KR	Kissing You(키싱유) 이세계아이돌 COVER [소녀시대 - Kissing You(키싱유)] (ISEGYE IDOL)	윌타버스 WAKTAVERSE	1093916	3
MX	HAPPY WHEELS, PERO EN 2024 !! - Episodio 28 Fernanfloo	Fernanfloo	7227426	1
MX	ENTRE AL TUNEL DE ALFA EN ROBLOX!	FedeGames	5938537	2
MX	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	3
US	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1
US	If my viewers break my secret rule, I ban them	DougDoug	2988844	2
US	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3

- **Number of Distinct Videos with a Title containing the word “BTS” (case insensitive):**

With 468 unique films featuring “BTS” in the title - much more than any other region – South Korea (KR) leads. Having 288 unique videos, India (IN) comes in second and demonstrates a significant amount of BTS-related content. The United States (US) trails closely after with 268 videos. Containing 262 videos, Canada (CA) has a sizable fan base for BTS-related content. Mexico (MX) exhibits noteworthy attention as well, with 254 unique videos. There are 251 videos from Japan (JP), demonstrating BTS’s significant presence there as well. Germany (DE) is the next country with 242 unique videos. 223 different videos from Great Britain (GB) demonstrate the popularity of BTS in the country. Following with 186 different videos is Brazil (BR). The region with the lowest ranking is France (FR), with 167 distinct videos featuring “BTS”.

Figure 6: Bar Chart of Regions (vertically) vs Distinct Video Count with word “BTS”



- The most viewed video for each country, year, and month and only for the year 2024, and its likes_ratio (defined as the percentage of likes against view_count):

The top popular videos for each country by month in 2024 have been identified through analysis. The percentage of likes compared to the view count, rounded to two decimal places, was used to compute the 'likes_ratio'. Month and country were used to sort the results. Across several regions and months in 2024, MrBeast is the most viewed content creator with a variety of films featuring challenges and survival themes. In January 2024, League of Legends (Riot Games) makes a significant appearance in a number of nations, especially with the “Still Here | Season 2024 Cinematic” video. In April 2024, Discord is also widely used, especially in the US, DE, and GB. While the 'likes_ratio' changes from video to video, it often stays between 3.0 and 4.0 for MrBeast videos, suggesting a strong engagement rate in comparison to views.

Figure 7: most viewed video for each country in 2024 and its 'likes_ratio'

COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO
BR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	139504939	3.20
CA	2024-01-01	Still Here Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
DE	2024-01-01	Still Here Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
FR	2024-01-01	Still Here Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
GB	2024-01-01	Still Here Season 2024 Cinematic - League of Legends (ft. Forts, Tiffany Aris, and 2WEI)	League of Legends	Gaming	104159411	1.69
IN	2024-01-01	Protect \$500,000 Keep It!	MrBeast	Entertainment	85458562	4.21
JP	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
KR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	143955997	3.16
MX	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22
US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	16623421	6.73
BR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	126846652	3.54
CA	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	119170728	3.66
DE	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
FR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
GB	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	114978689	3.72
IN	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	109797680	3.81
JP	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	130451673	3.49
KR	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	139979994	3.34
MX	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	134662016	3.42
US	2024-02-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	144360336	3.29
BR	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	127664536	3.75
CA	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	124900354	3.79
DE	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	117665895	3.91
FR	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	121793723	3.84
GB	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	124900354	3.79
IN	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	97224959	4.33
JP	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	132125566	3.89
KR	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	140144693	3.58
MX	2024-03-01	I Survived 7 Days In An Abandoned City	MrBeast	Entertainment	142313377	3.56
US	2024-03-01	Face Your Biggest Fear To Win \$800,000	MrBeast	Entertainment	169649341	3.00
BR	2024-04-01	7 Days Stranded On An Island	MrBeast	Entertainment	108502278	4.25
CA	2024-04-01	7 Days Stranded On An Island	MrBeast	Entertainment	108502278	4.25
DE	2024-04-01	Discord Loot Boxes are here.	Discord	Entertainment	1407643634	0.01
FR	2024-04-01	Pushpa 2 The Rule Teaser Allu Arjun Sukumar Rashmika Mendanna Fahadh Faasil DSP	Mythri Movie Makers	Entertainment	112497953	1.45
GB	2024-04-01	Discord Loot Boxes are here.	Discord	Entertainment	1406329649	0.01
IN	2024-04-01	Pushpa 2 The Rule Teaser Allu Arjun Sukumar Rashmika Mendanna Fahadh Faasil DSP	Mythri Movie Makers	Entertainment	112497953	1.45
JP	2024-04-01	BABYMONSTER - "SHEESH" MV	BABYMONSTER	People & Blogs	117214985	2.07
KR	2024-04-01	BABYMONSTER - "SHEESH" MV	BABYMONSTER	People & Blogs	117214985	2.07
MX	2024-04-01	Discord Loot Boxes are here.	Discord	Entertainment	1406329649	0.01
US	2024-04-01	Discord Loot Boxes are here.	Discord	Entertainment	1407643634	0.01

- **`CATEGORY\_TITLES` with the most distinct videos for each country and its percentage out of the total distinct number of videos of that country from the year 2022.**

Categories with the most distinct videos for each country in 2022, showing the total number of videos for that category have been analyzed, including the total number of videos for the country, and the percentage that the category's videos represent in total. The results are ordered by category title and country.

Figure 8: most viewed video for each country in 2024 and its `likes\_ratio`

	COUNTRY	CATEGORY_TITLE	TOTAL_CATEGORY_VIDEO	TOTAL_COUNTRY_VIDEO	PERCENTAGE
1	BR	Entertainment	5417	23760	22.80
2	DE	Entertainment	7709	30719	25.10
3	FR	Entertainment	7548	32849	22.98
4	GB	Entertainment	5643	27855	20.26
5	IN	Entertainment	21281	50250	42.35
6	JP	Entertainment	5658	17627	32.10
7	KR	Entertainment	5122	15175	33.75
8	MX	Entertainment	4195	17532	23.93
9	CA	Gaming	6594	30869	21.36
10	US	Gaming	6226	28799	21.62

In most countries, entertainment dominates and makes up a significant portion of all distinct videos. With 50,250 unique videos, India (IN) leads all the countries and accounts for the largest percentage of all videos (42.35%). South Korea (KR) with 33.75% of the total, it follows closely behind. Japan (JP) with 17,627 videos or 32.10% of the total, demonstrates a significant presence as well. Germany (DE) contains 30719 videos, or 25.10%, are available. With 32,849 videos, France (FR) makes up 22.98% of the total. Brazil (BR) has 23,760 videos or 22.80% of the total. Mexico (MX), with 17,532 videos, or 23.93% of the total, has the fewest amount of entertainment videos. With 27855 videos, entertainment accounts for 20.26% in Great Britain (GB). With 28,799 unique videos, or 21.62% of the total, and 30869 game videos, or 21.36% of the total, in the US and Canada, respectively, gaming is the most popular category.

- **`CHANNEL\_TITLE` with the most distinct videos and its count:**

“Vijay Television” has been found to be the channel with the highest number of unique videos (2049).

4. Business Question

An in-depth analysis of several content categories – aside from “Music” and “Entertainment” – has been performed to choose the optimal strategy for launching a new YouTube channel.

- **Top Categories by Video Count and Engagement:** With 43,323 videos overall, the “Sports” category has the most distinct trending videos. However, the “Gaming” category stood out in terms of interaction metrics, such as likes and comments, having the most in all regions with 11,192,833 likes and 1,280,276 comments.
- **Category with the Highest View Count:** The category with the most view count was “Science & Technology,” with 183,275,166 views, showing that there was a lot of interest in this field. Comparing this category to others like “Gaming,” there were, nevertheless, fewer trending videos overall.
- **Composite Score Analysis:** Each category has a composite score that is determined by considering their view counts and presence in the trending list. With the highest composite score, the “Sports” category is the most prominent due to a combination of high view counts and a significant number of trending videos.
- **Cross-Country Trend Analysis:** The “Gaming” category proved to be the most consistent across most regions in terms of having the greatest engagement metrics and view count. Since user interactions, such as comments and likes, are the direct indicator of a video’s reach and popularity, this makes it an ideal choice for a channel launch that aims to be the top trending category worldwide.

Although the “Sports” category has the highest composite score and the most number of distinct trending videos, the “Gaming” category is better for launching a new YouTube channel due to its steady popularity and high level of involvement in several different regions.

5. Issues Faced and Their Solutions

- Defining CSV File Formats
 - Issue: I struggled with how to manage null values efficiently and provide the required format options, in addition to `SKIP_HEADER` and `FIELD_DELIMITER` when setting the file format for the CSV files.
 - Solution: To efficiently manage potential null values and formatting options, I developed a robust CSV file format that contained options like `FIELD_OPTIONALLY_ENCLOSED_BY`, `TRIM_SPACE`, and `NULL_IF`:

Figure 9: SQL Code block of CSV file format

```
-- CSV File Format
CREATE OR REPLACE FILE FORMAT csv_file_format
TYPE = 'CSV'
FIELD_DELIMITER = ','
SKIP_HEADER = 1
FIELD_OPTIONALLY_ENCLOSED_BY = ''
TRIM_SPACE = TRUE
NULL_IF = ('\N', 'NULL', '');
```

- Error in TO_TIMESTAMP_NTZ Function

- Issue: The TO_TIMESTAMP_NTZ function gave an error message stating that there were too many arguments supplied.

Figure 10: Error in timestamp while creating an external table for trending data

```
Error: too many arguments for function
[TO_TIMESTAMP_NTZ(GET(EX_TABLE_YOUTUBE_TRENDING.VALUE,
'c3'), 'YYYY-MM-DDTHH24:MI:SSZ')] expected 1, got 2 (line 28)
```

- Solution: Passing a second parameter to TO_TIMESTAMP_NTZ, which only takes one argument, caused the issue. I fixed the issue with the help of ChatGPT, by making sure the timestamp conversion was performed correctly.
- Incorrect SQL Query for Duplicate Video Records
 - Issue: The initial query for retaining the counts for the highest view_count among duplicates records, was partitioned incorrectly.
 - Solution: With the help of ChatGPT, I made adjustments to the original query to make sure that the highest number of view_count will be the record to keep:

Figure 11: “table_youtube_duplicates” containing only the “bad” duplicates

```
CREATE OR REPLACE TABLE table_youtube_duplicates AS
WITH most_viewed AS (
  SELECT
    *,
    ROW_NUMBER() OVER (
      PARTITION BY VIDEO_ID, TRENDING_DATE, COUNTRY
      ORDER BY VIEW_COUNT DESC
    ) AS row_no
  FROM
    table_youtube_final
)
SELECT
  *
FROM
  most_viewed
WHERE
  row_no > 1;
```

- Unsupported Subquery Type in `UPDATE` Statement

- Issue: I received the following problem when trying to update the table `table\_youtube\_final` with values from `table\_youtube\_category`:
“Unsupported subquery type cannot be evaluated.”

- Solution: To fix this, I adjusted the update query to use a join – which Snowflake supports – instead of a subquery to carry out the update operation:

Figure 12: Updated query to replace the NULL values in `table\_youtube\_final`

```
UPDATE table_youtube_final yt
SET yt.CATEGORY_TITLE = yc.CATEGORY_TITLE
FROM table_youtube_category yc
WHERE yt.CATEGORY_TITLE IS NULL
AND yt.CATEGORYID = yc.CATEGORYID;
```

- Using the TRUNC Function to Round
 - Issue: The figures were not rounded to the correct amount when truncating them with the TRUNC function to calculate LIKES_RATIO (1.686, for example, was truncated to 1.68 instead of rounding to 1.69).
 - Solution: To get the proper rounded values, I swapped out the TRUNC function for the ROUND function.

6. Conclusion

Over a considerable amount of period and in several different countries, the analysis effectively offered insights into YouTube video trends. Efficient handling of data and query execution were made possible by utilizing Snowflake's Data Lakehouse architecture. The results can be applied to improve YouTube audience targeting and content production tactics.